Task 6 – Patch Management

Cyber Security Internship

Prepared as an internship task submission

1. Introduction

Patch management is the process of keeping operating systems and applications updated. Security patches are particularly important because they can fix vulnerabilities that attackers may otherwise exploit.

2. Why Patch Management Matters

A system can become vulnerable simply because an important update was never installed. Regular patching reduces the time a known vulnerability remains exposed and can also improve stability and compatibility.

3. A Practical Patch Management Process

• Maintain an inventory of computers, servers and applications.

• Identify available updates and security patches.

• Prioritize patches based on severity, exposure and business importance.

• Test important patches where possible before broad deployment.

• Deploy the approved updates.

• Verify that updates were installed successfully.

• Record the work and continue monitoring for new patches.

4. Common Challenges

Patching sounds simple, but organizations can face compatibility problems, downtime requirements, legacy systems and limited staff or resources. This is why patching is normally handled as a planned process rather than simply installing every update without checking its impact.

5. Prioritizing Security Patches

Not every update has the same urgency. A critical vulnerability affecting an internet-facing system may deserve immediate attention, while a lower-risk update on an isolated system may be scheduled later. Risk, exposure and business impact should guide the decision.

6. Backups and Recovery

Before major changes, appropriate backups and recovery procedures are important. If an update causes an unexpected problem, a tested recovery process can reduce downtime and data loss.

7. Automation

Organizations can use tools such as Windows Update, WSUS, Microsoft Intune and Linux package-management systems to help manage updates. Automation can reduce repetitive work, but it should still be monitored.

8. Best Practices

• Keep an accurate asset inventory.

• Use a regular patching schedule.

• Prioritize security-critical vulnerabilities.

• Test important updates when practical.

• Keep backups and recovery procedures available.

• Verify successful installation.

• Document exceptions when systems cannot be patched immediately.

9. Personal Takeaway

The main lesson for me is that patch management is not just clicking an update button. It is a continuing security process involving identification, prioritization, testing, deployment and verification.

10. Conclusion

A consistent patch management process reduces exposure to known vulnerabilities and helps organizations maintain more reliable systems. It should be treated as a normal part of cybersecurity rather than something done only after an incident.